From RAGs to Riches!

END-TO-END RAG WITH WEAVIATE



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datacamp

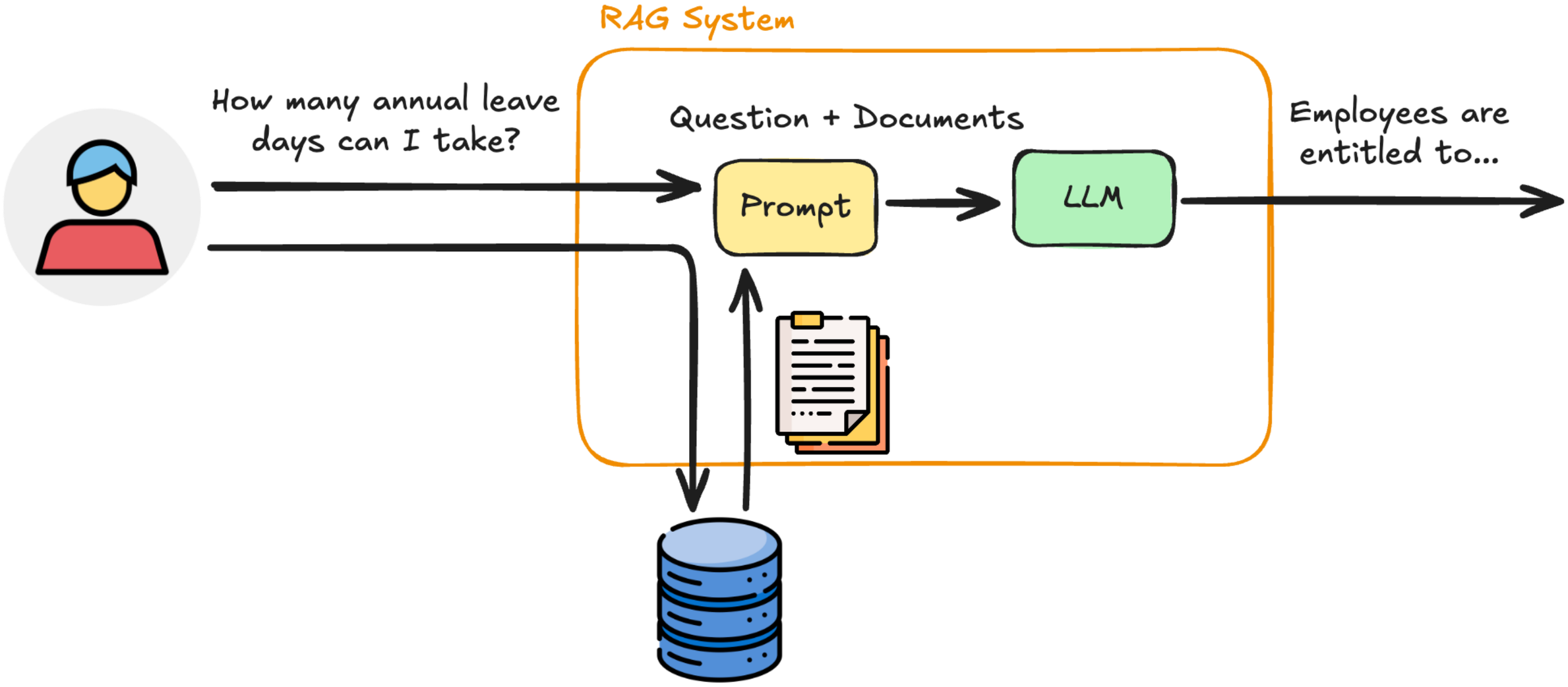
Meet your instructor!



Senior Developer Educator,

Weaviate

The journey ahead





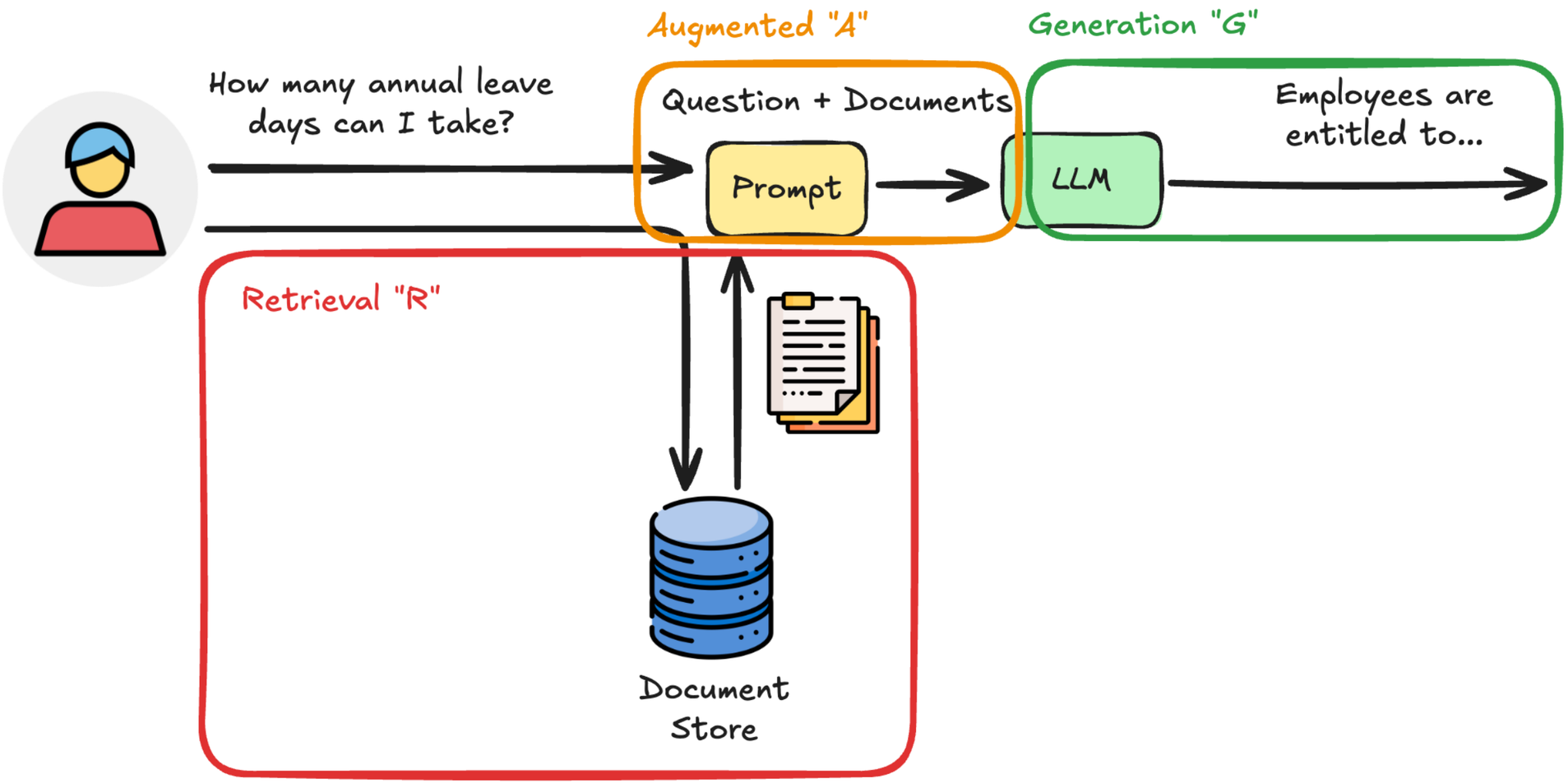
Weaviate

→ Open-source

→ Cloud-native

→ Vector database

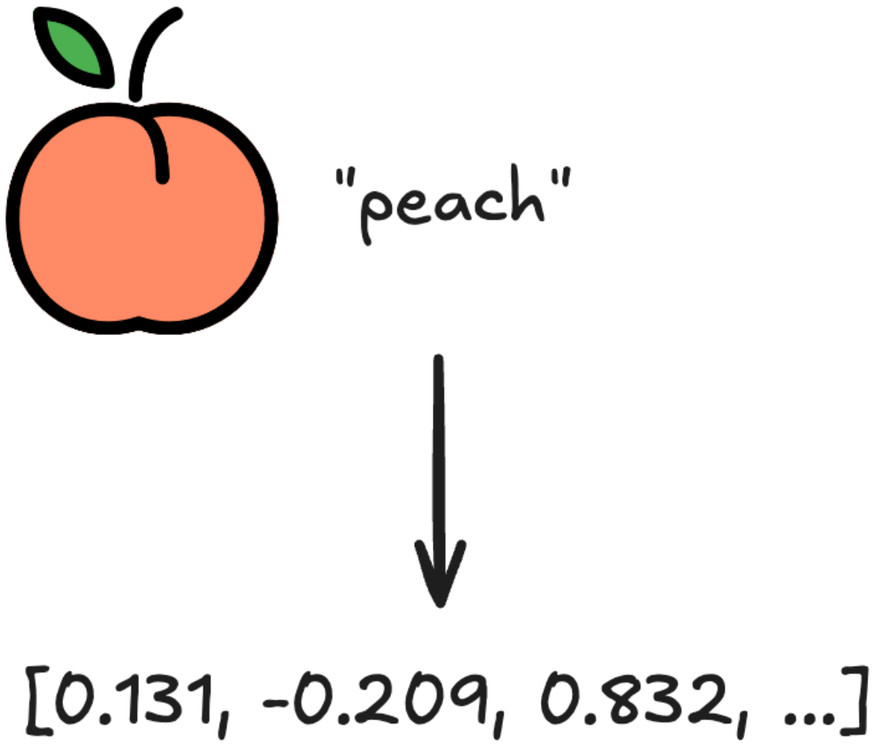
RAG in a nutshell



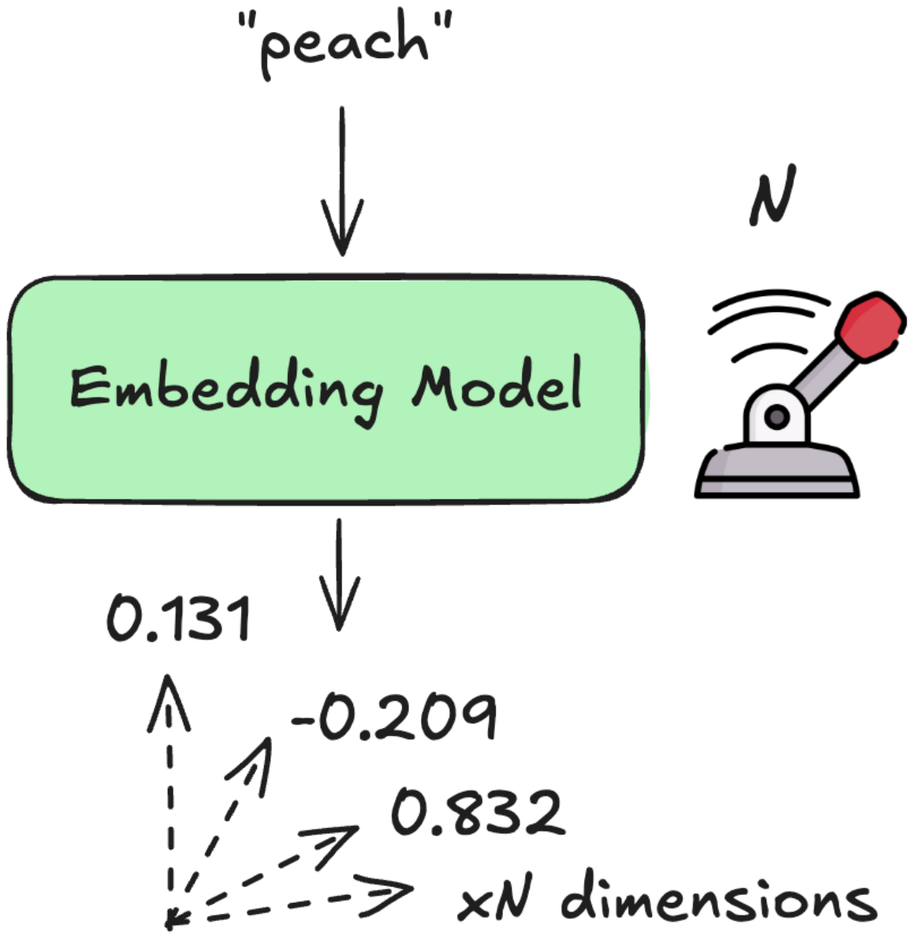
Text embeddings

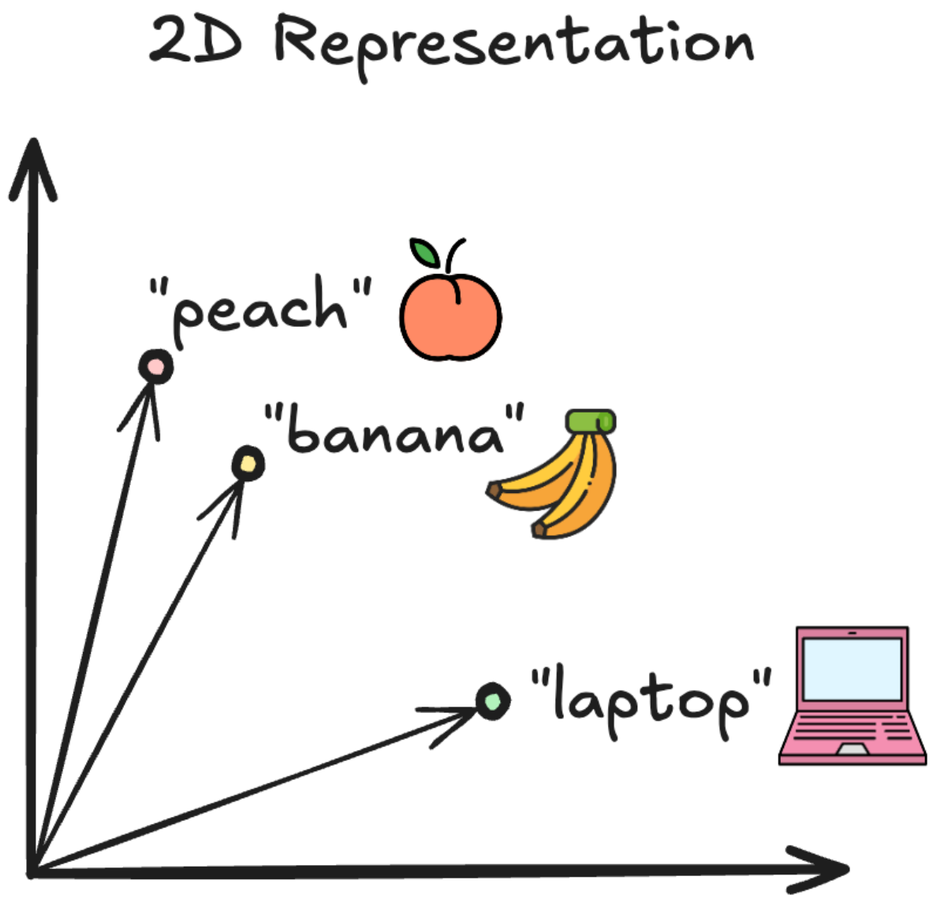
- **Chapter 3** → image embeddings
- **Text embeddings:** encode semantic meaning of text numerically

- Can be used to compare the *semantic similarity* between text

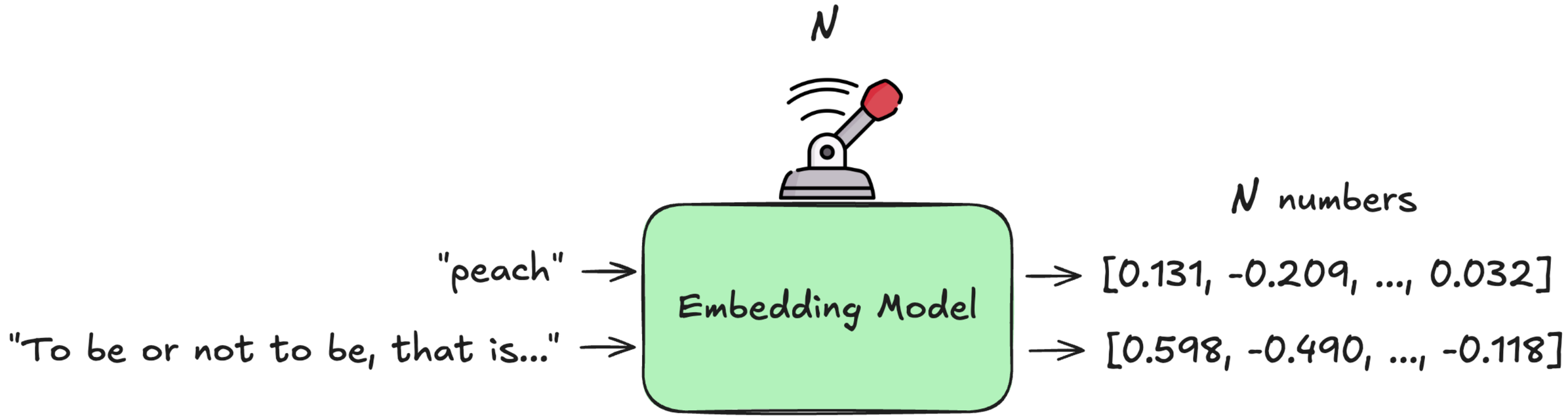


Visualizing text embeddings

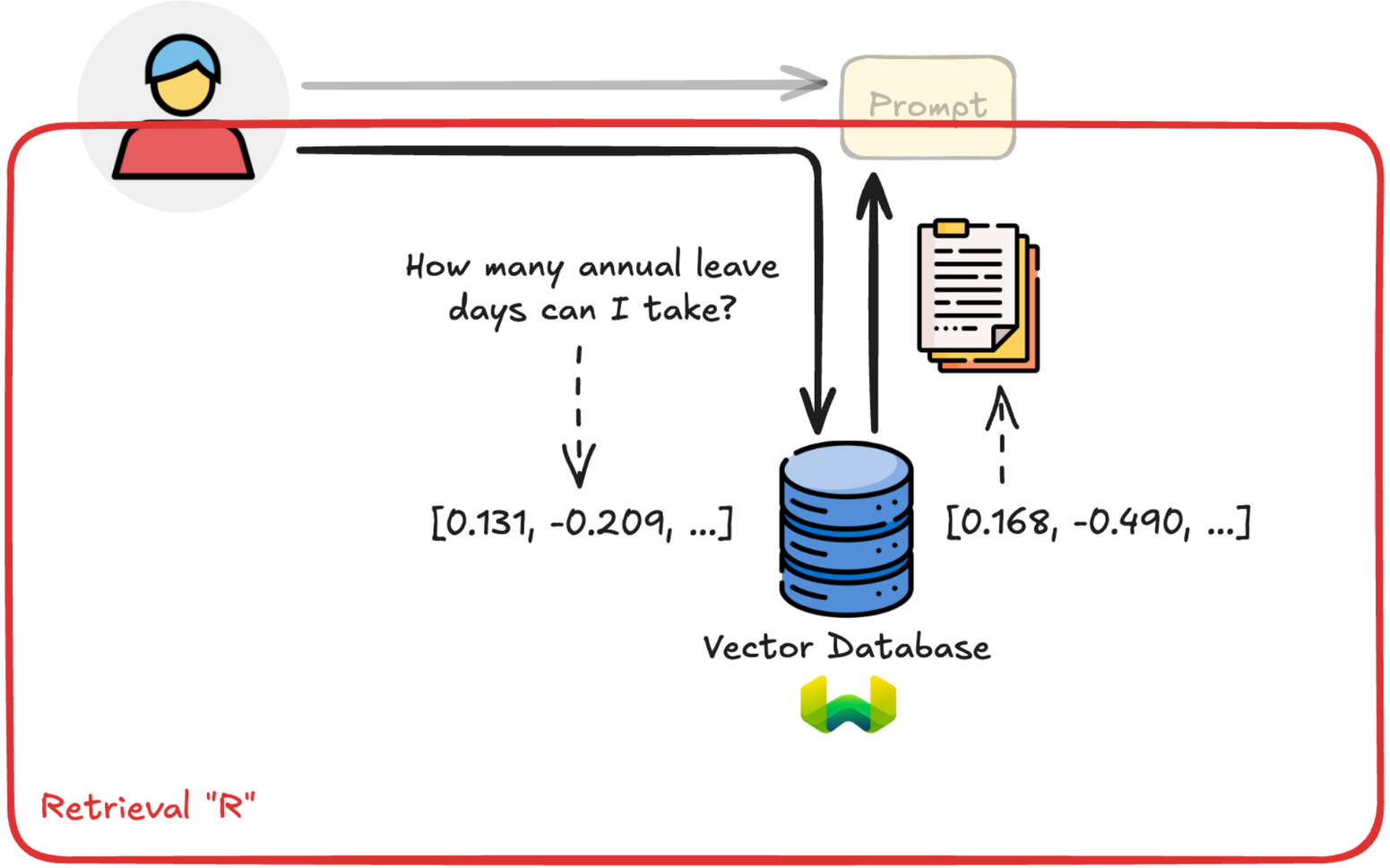




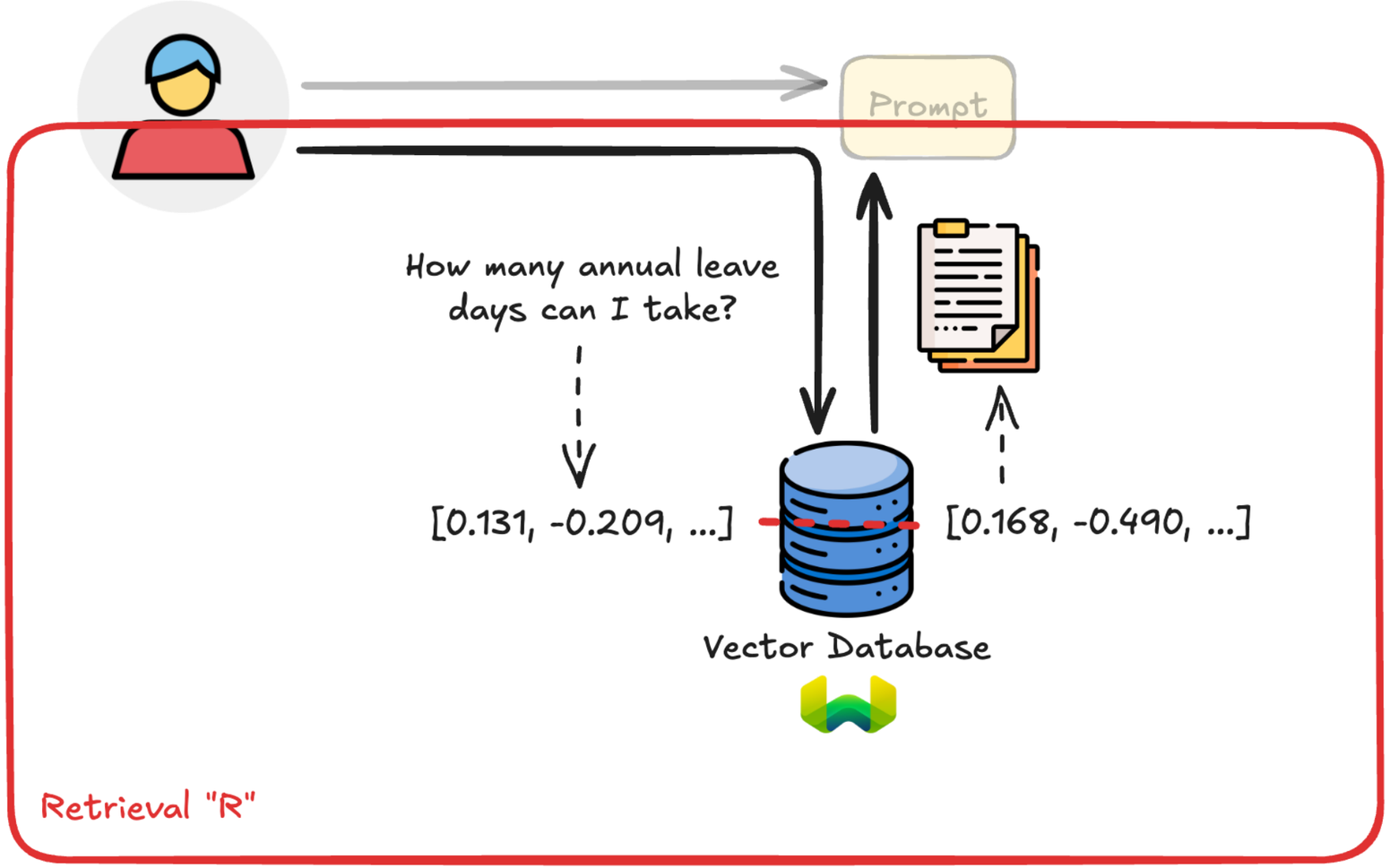
Consistent vector length



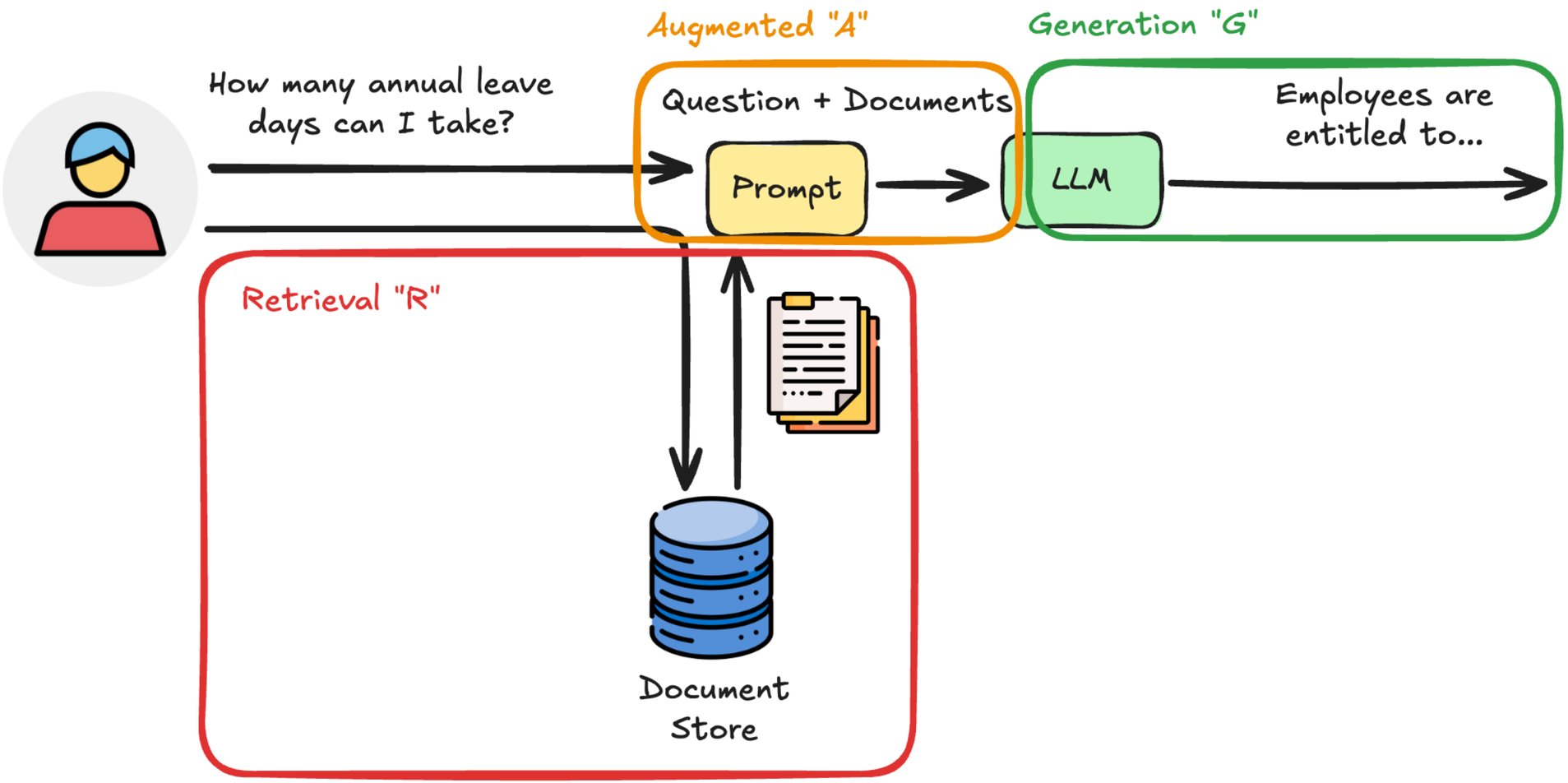
Embeddings and retrieval



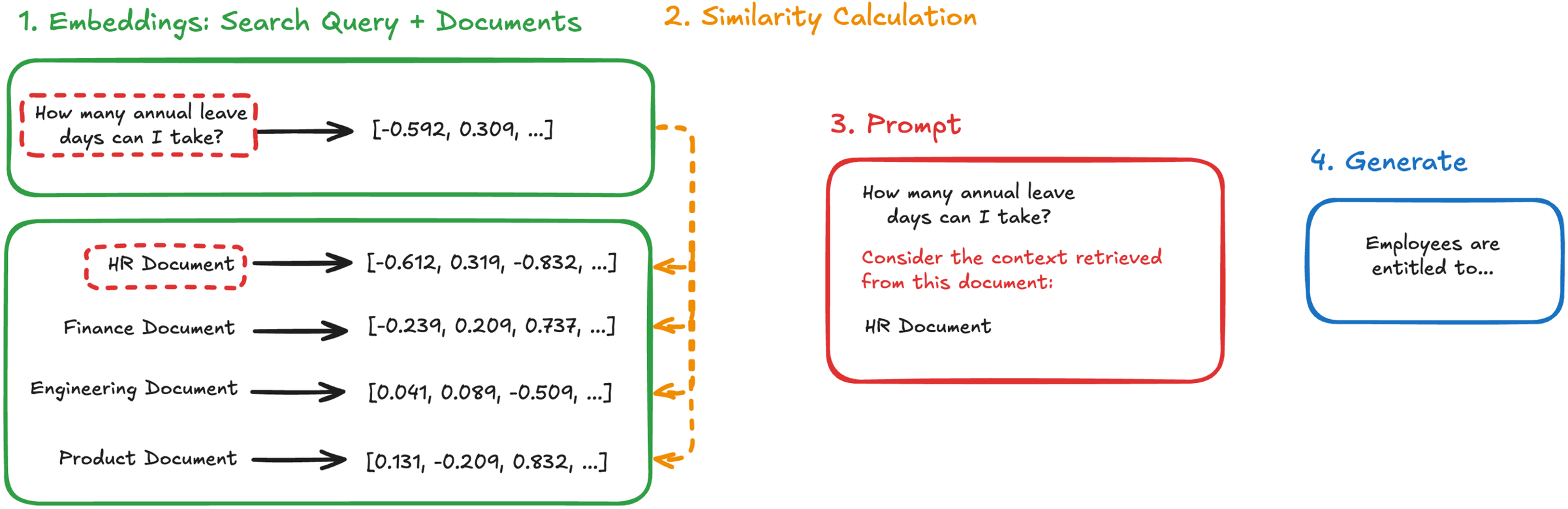
Embeddings and retrieval



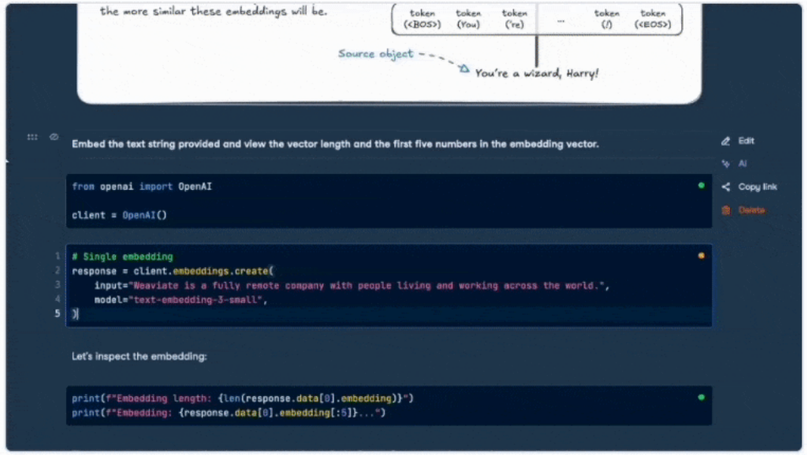
"R" is for Retrieval



Coming up!


1. Embeddings: Search Query + Documents
2. Similarity Calculation
How many annual leave days can I take?
[-0.592, 0.309, ...]
HR Document
[-0.612, 0.319, -0.832, ...]
Finance Document
[-0.239, 0.209, 0.737, ...]
Engineering Document
[0.041, 0.089, -0.509, ...]
Product Document
[0.131, -0.209, 0.832, ...]
3. Prompt
How many annual leave days can I take?
Consider the context retrieved from this document:
HR Document
4. Generate
Employees are entitled to...

Video



Instructions 100XP

- Embed the text string provided and view its length and the first five numbers in the embedding vector.
- Embed the strings in the `texts` list in a single embedding API request, and compare the length and numbers of each embedding vector.
- Embed the `texts` list again, but this time, change the model output dimensions to `512`.

Take Hint (-30 XP)

File Environment

Embed the text string provided and view the vector length and the first five numbers in the embedding vector.

```
from openai import OpenAI

client = OpenAI()
```

```
# Single embedding
response = client.____.____(
    ____="Weaviate is a fully remote company with people living and working across the world.",
    model="text-embedding-3-small",
)
```

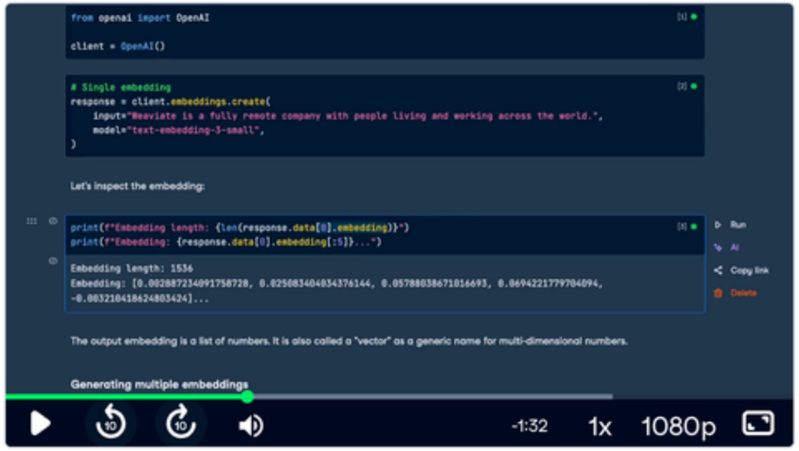
Let's inspect the embedding:

```
print(f"Embedding length: {len(response.data[0].embedding)}")
print(f"Embedding: {response.data[0].embedding[:5]}...")
```

The output embedding is a list of numbers. It is also called a "vector" as a generic name for multi-dimensional numbers.

Run All Submit Answer

Video

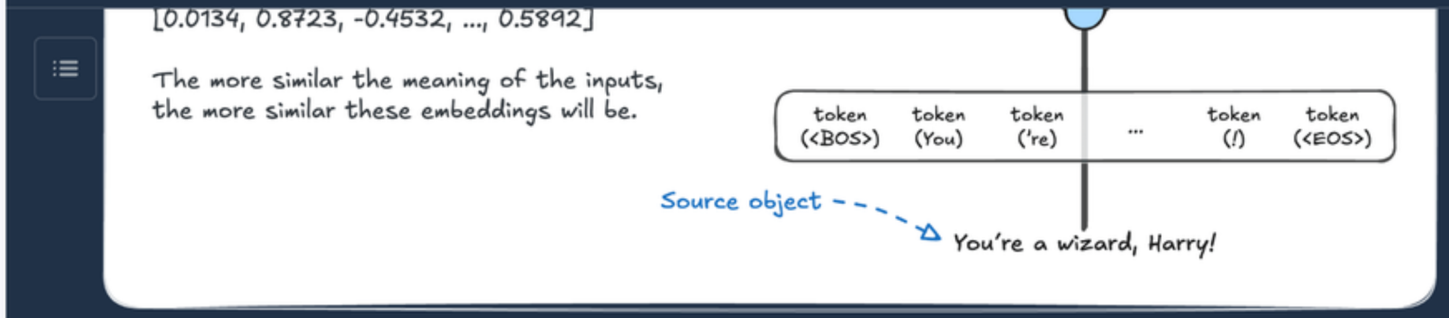


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Take Hint (-30 XP)

File Environment Modified by James Chapman, 20 min ago



Embed the text string provided and view the vector length and the first five numbers in the embedding vector.

```
from openai import OpenAI                                    [1]

client = OpenAI()
```

```
# Single embedding                                           [2]
response = client.embeddings.create(
    input="Weaviate is a fully remote company with people living and working across the world.",
    model="text-embedding-3-small",
)
```

Let's inspect the embedding:

Run All Submit Answer

Let's practice!

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Text generation with LLMs

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Let's practice!

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Creating text embeddings

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Let's practice!

END-TO-END RAG WITH WEAVIATE

Comparing text embeddings

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Let's practice!

END-TO-END RAG WITH WEAVIATE

Putting together the "R", "A", "G"

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Let's practice!

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